

Digestive system

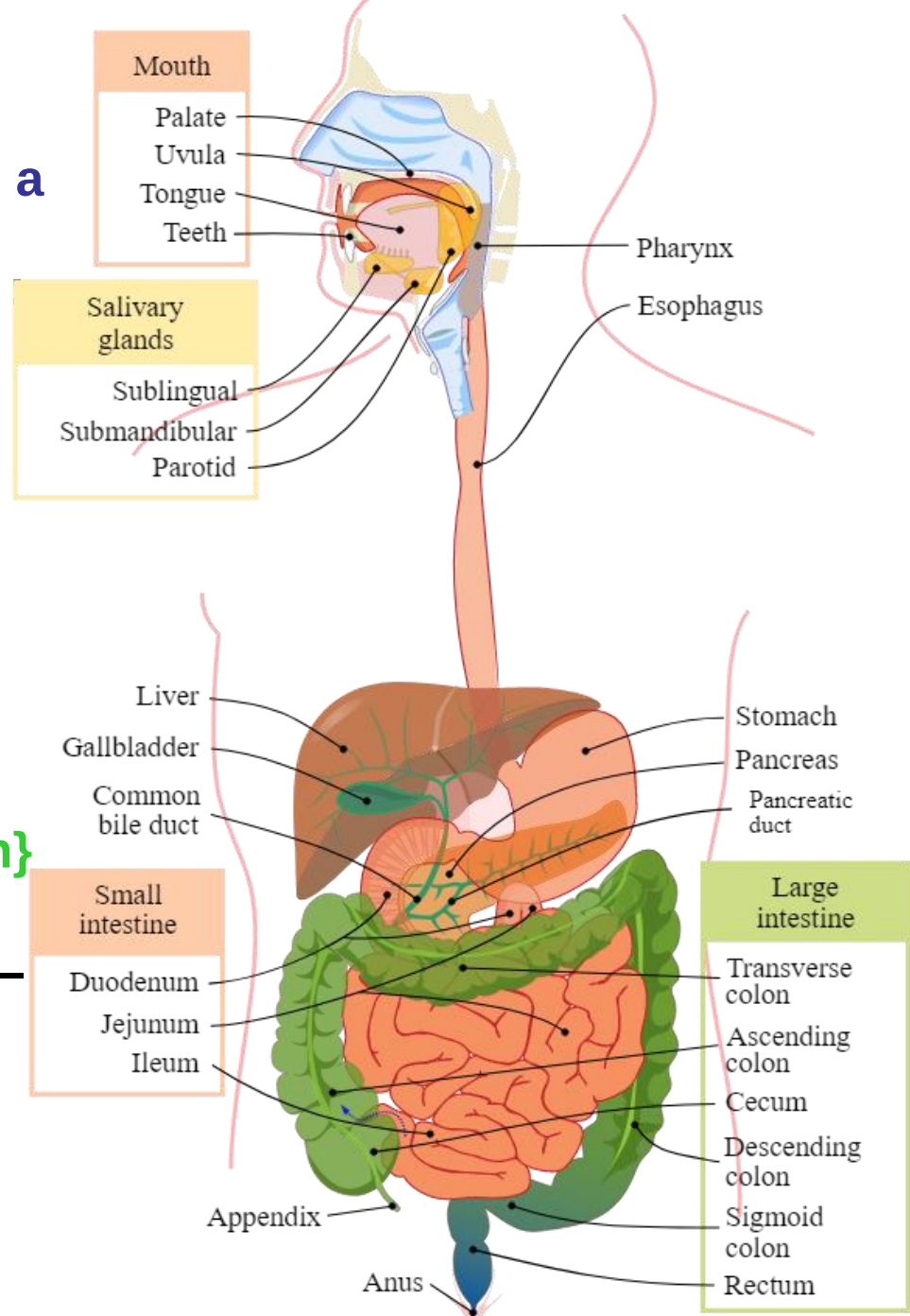
**The basic layers of the wall of
gastrointestinal tract**

LG1

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- **The digestive tract :**
- **is a series of organs forming a long muscular tube, the organs include the following**
- **Oral cavity**
- **pharynx**
- **Esophagus**
- **Stomach**
- **Small intestine (duodenum, jejunum & ileum)**
- **Large intestine (cecum, appendix, {ascending-transverse-descending colon}, rectum & anal canal)**
- **Accessory digestive organs – teeth, tongue, gallbladder, salivary glands, liver, and pancreas**



Histology of the Alimentary Canal

- From esophagus to the anal canal the walls of the GI tract have the same four tunics:

1-Mucosa

a-epithelial lining

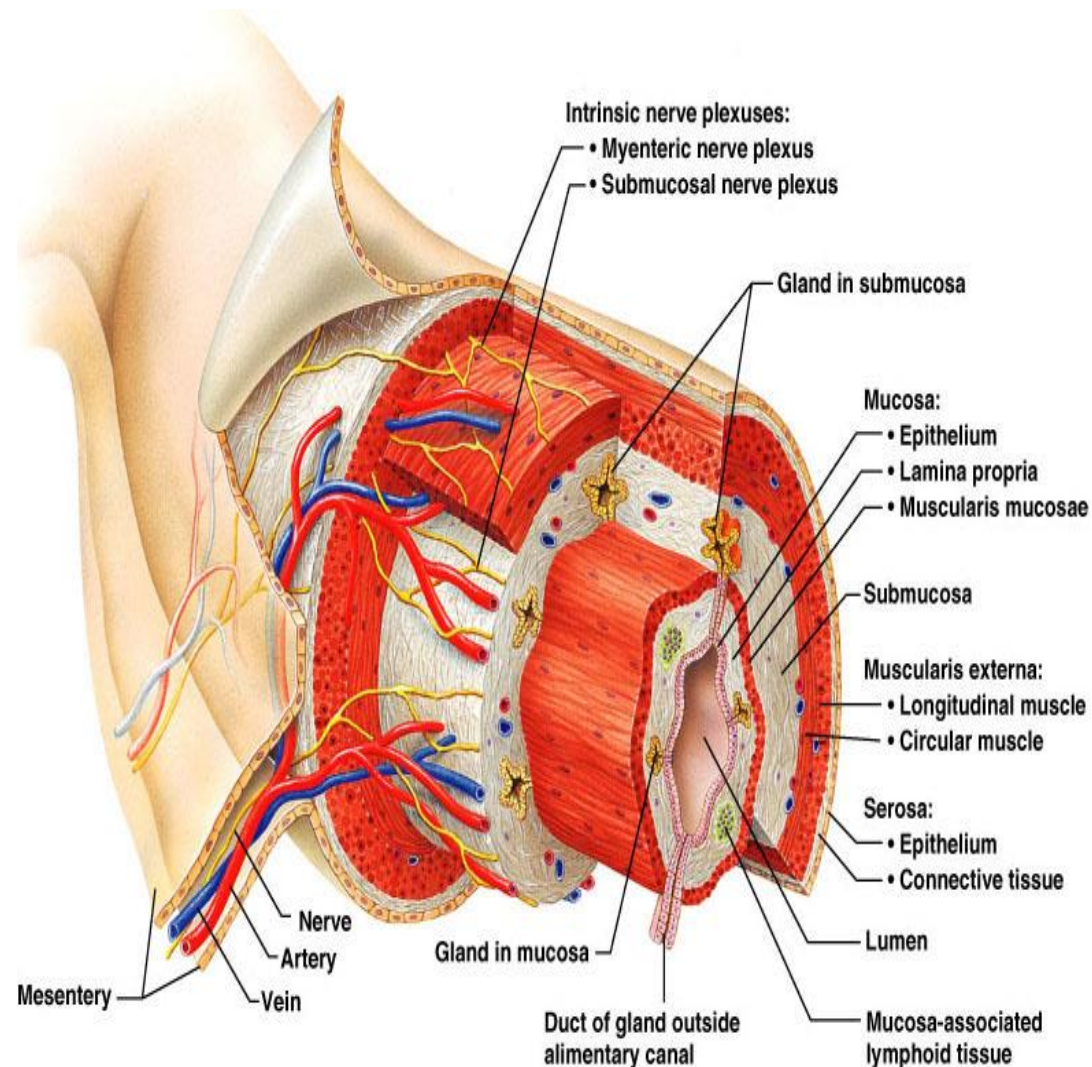
b-lamina propria

c-muscularis mucosa

2-Submucosa

3-Muscularis externa

4-Serosa, or adventitia



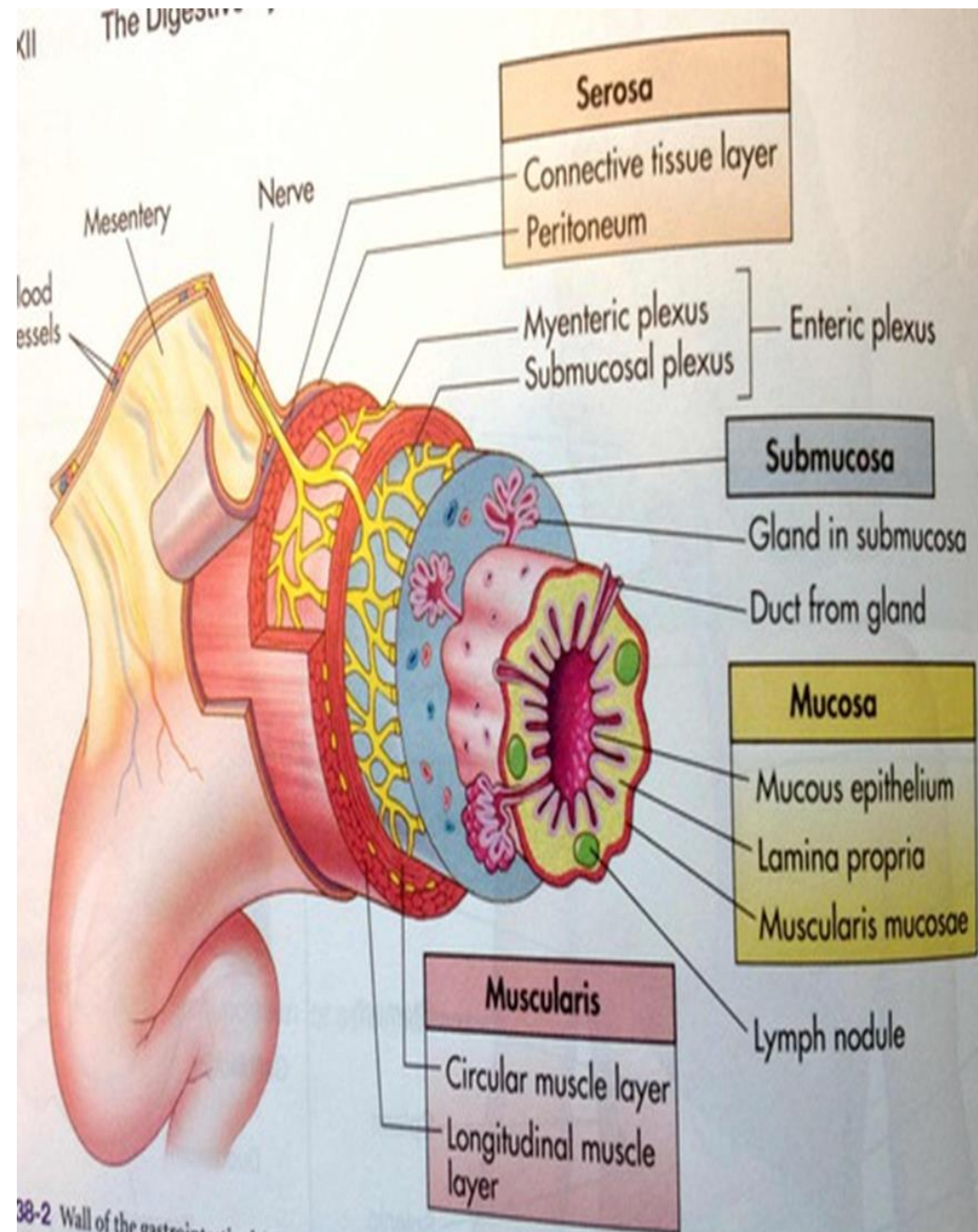
Mucosa

- The main functions of the mucosa
 - Secretion
 - absorption
 - Protection
- Consists of three layers:
 - a lining epithelium
 - lamina propria
 - muscularis mucosae

Mucosa

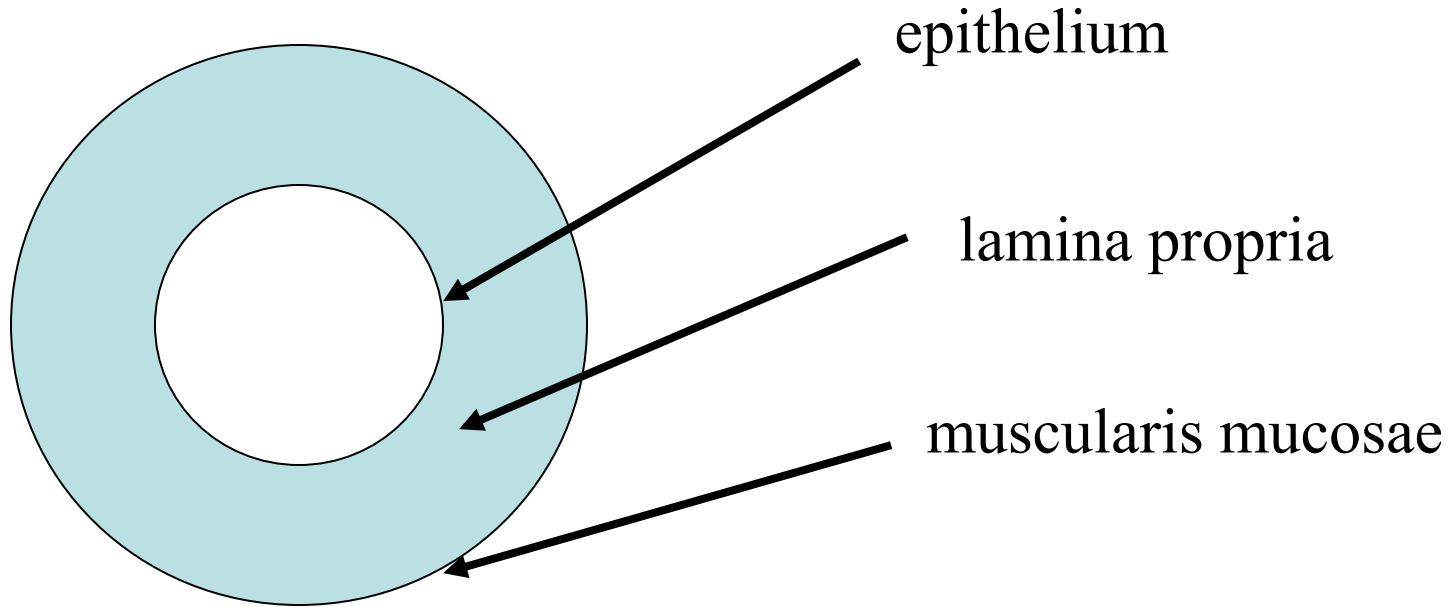
- **Epithelial** The epithelial lining of the mucosa varies from stratified squamous that able to resist abrasion, to tall columnar with a brush border, suitable for absorption
- **Lamina Propria** : The lamina propria is a loose connective tissue layer that supports the epithelium, contain blood vessels & lymphatic tissue , also contain glands open onto the epithelium.

Muscularis mucosae – is an incomplete layer of smooth muscle cells between mucosa & submucosa.



Characteristics of the GI Tube

Mucosa



Epithelium: stratified squamous --> simple columnar --> stratified squamous

esophagus

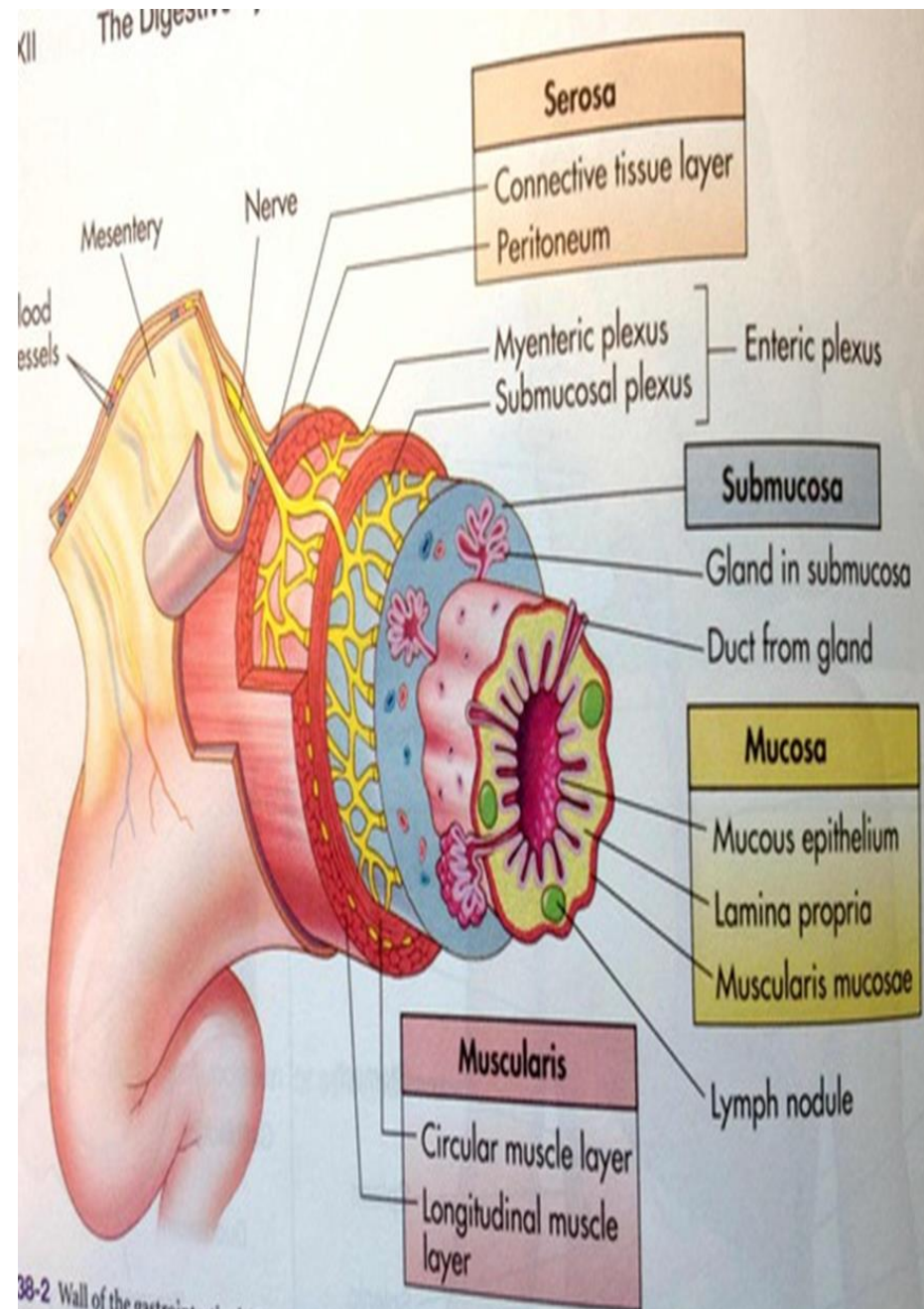
stomach --> rectum

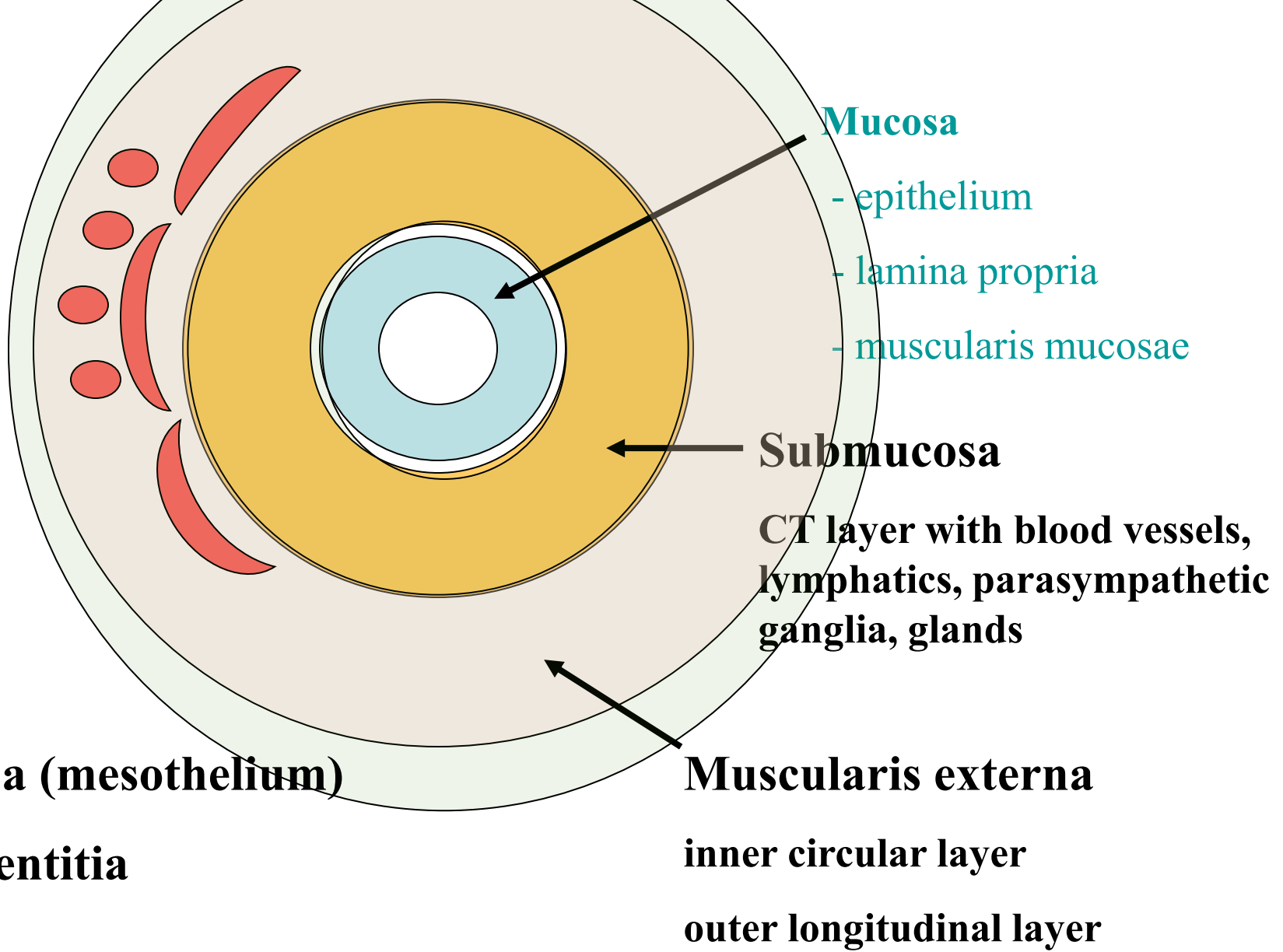
anus

Lamina propria: loose CT - rich in cells, capillaries, lymphatic vessels

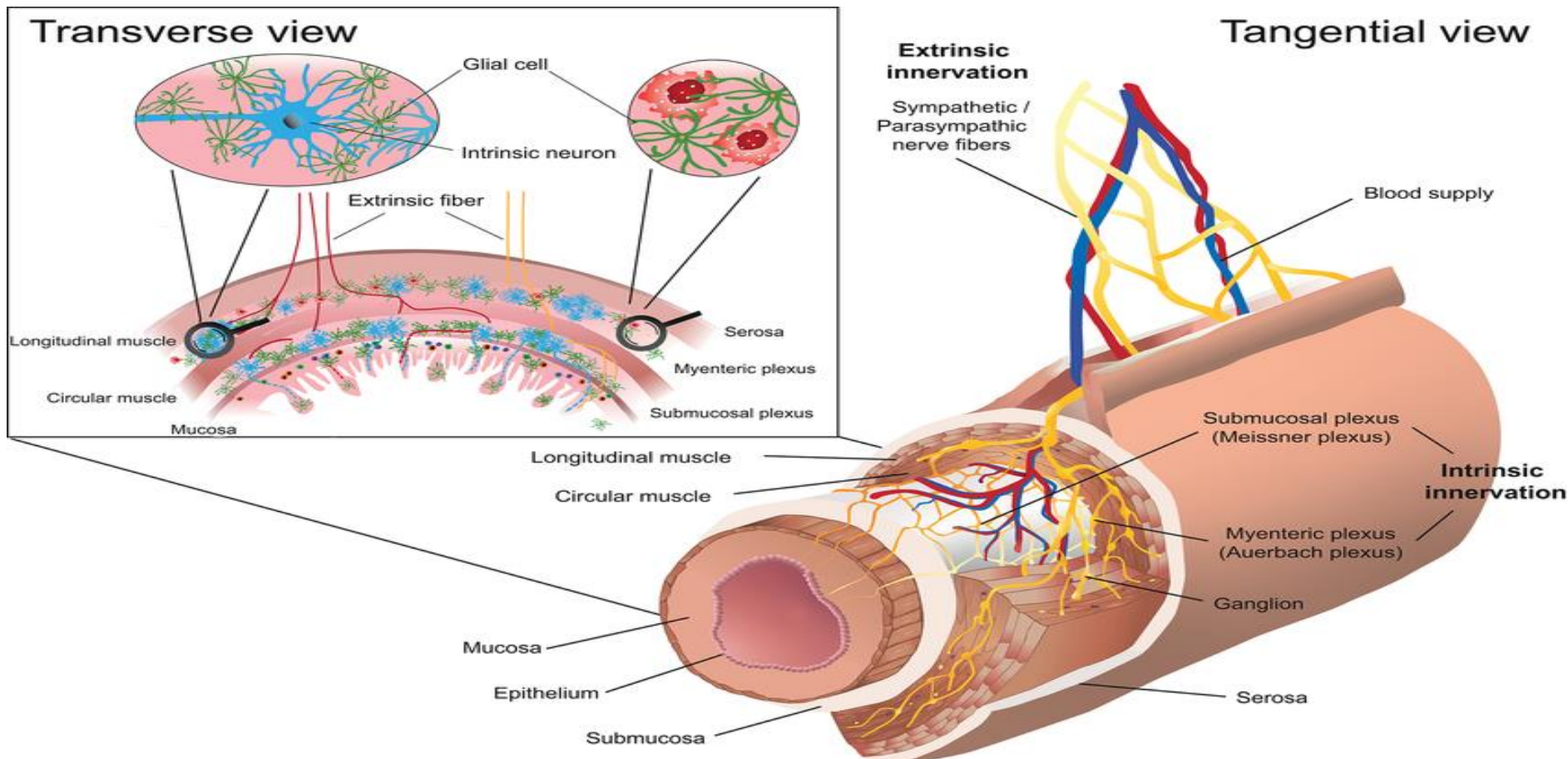
Muscularis mucosae: smooth muscle, two layers

- **Submucosa** – dense connective tissue containing blood and lymphatic vessels. Meissner's plexus is formed by nerve fibers & ganglion cells & controls glandular secretion.
- **Muscularis externa** – is the muscle coat of the gut tube. it is usually formed of an inner circular & outer longitudinal layers of smooth muscle.
- In some parts striated muscle replaces the smooth muscle, & in other 3 layers of smooth muscle may be present as in stomach. Auerbach's plexus (Myenteric plexus) of nerve fibers & ganglion cells found between the muscle coat & controls their function.



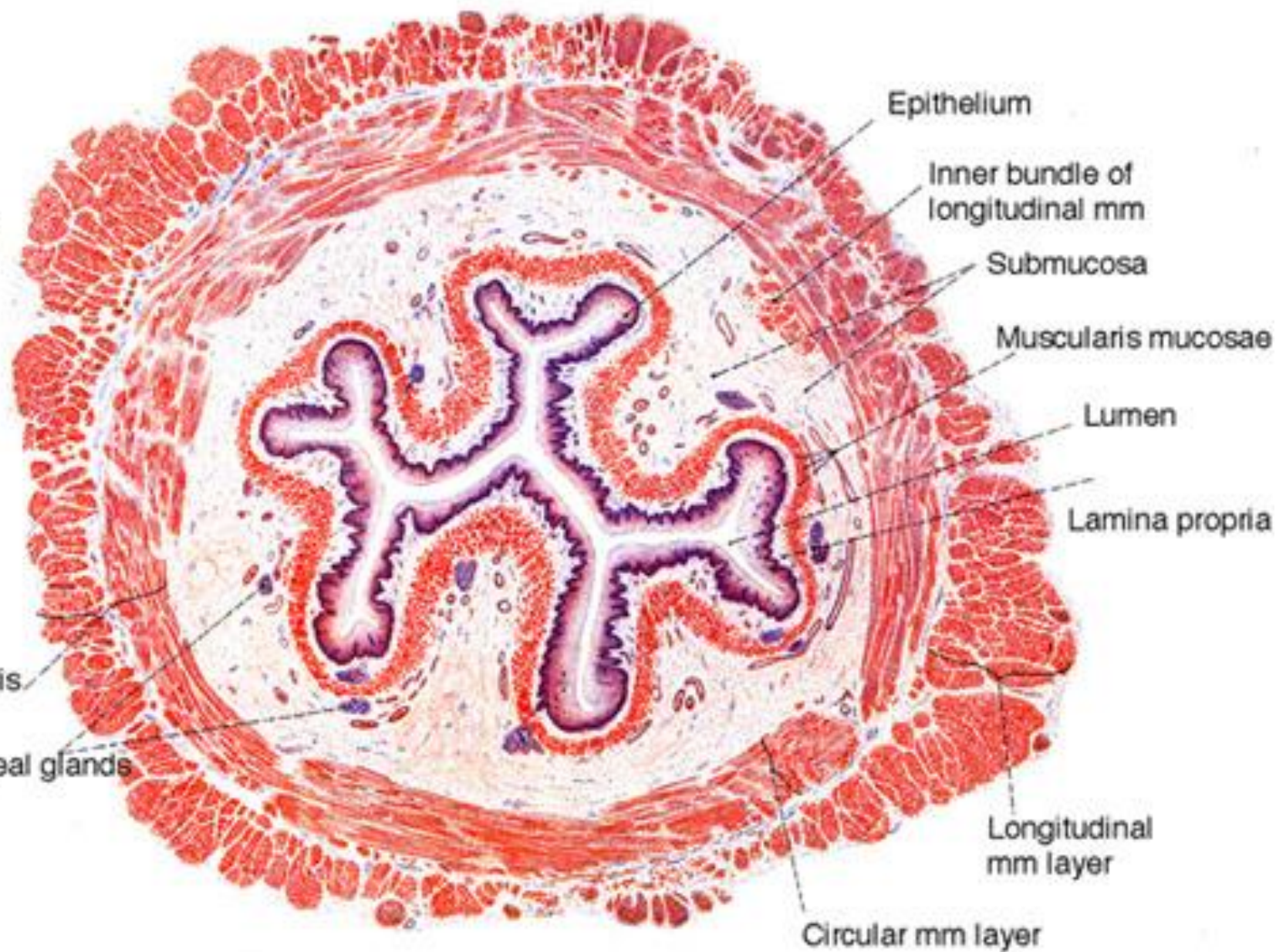


- **Serosa** –is connective tissue covered by a layer of simple squamous epithelium known as the mesothelium.
- Is outermost layer, form the visceral peritoneum
- the tissue which covers the organs of the peritoneal cavity.
- In some organs or part of organs covered by adventitia.

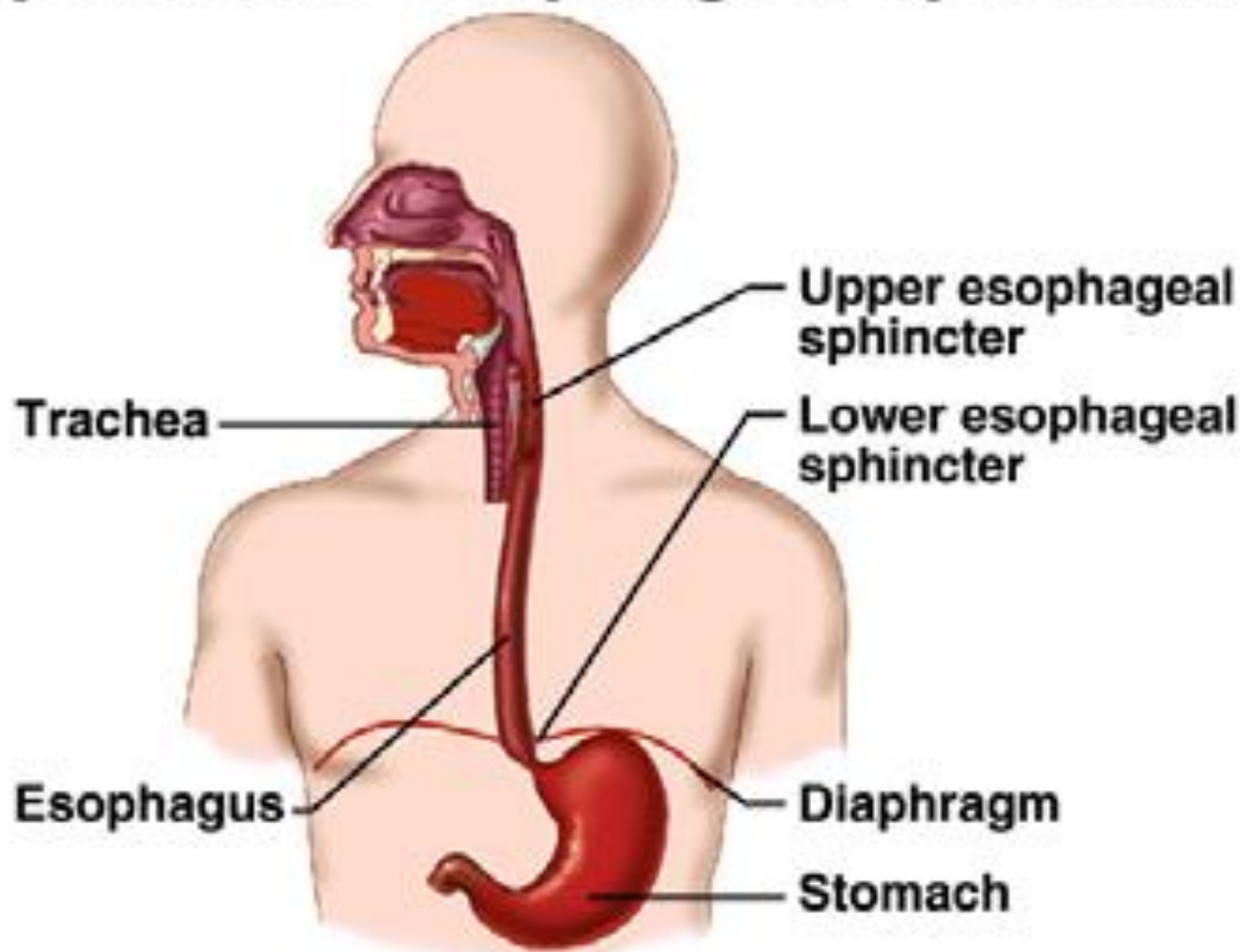


Esophagus

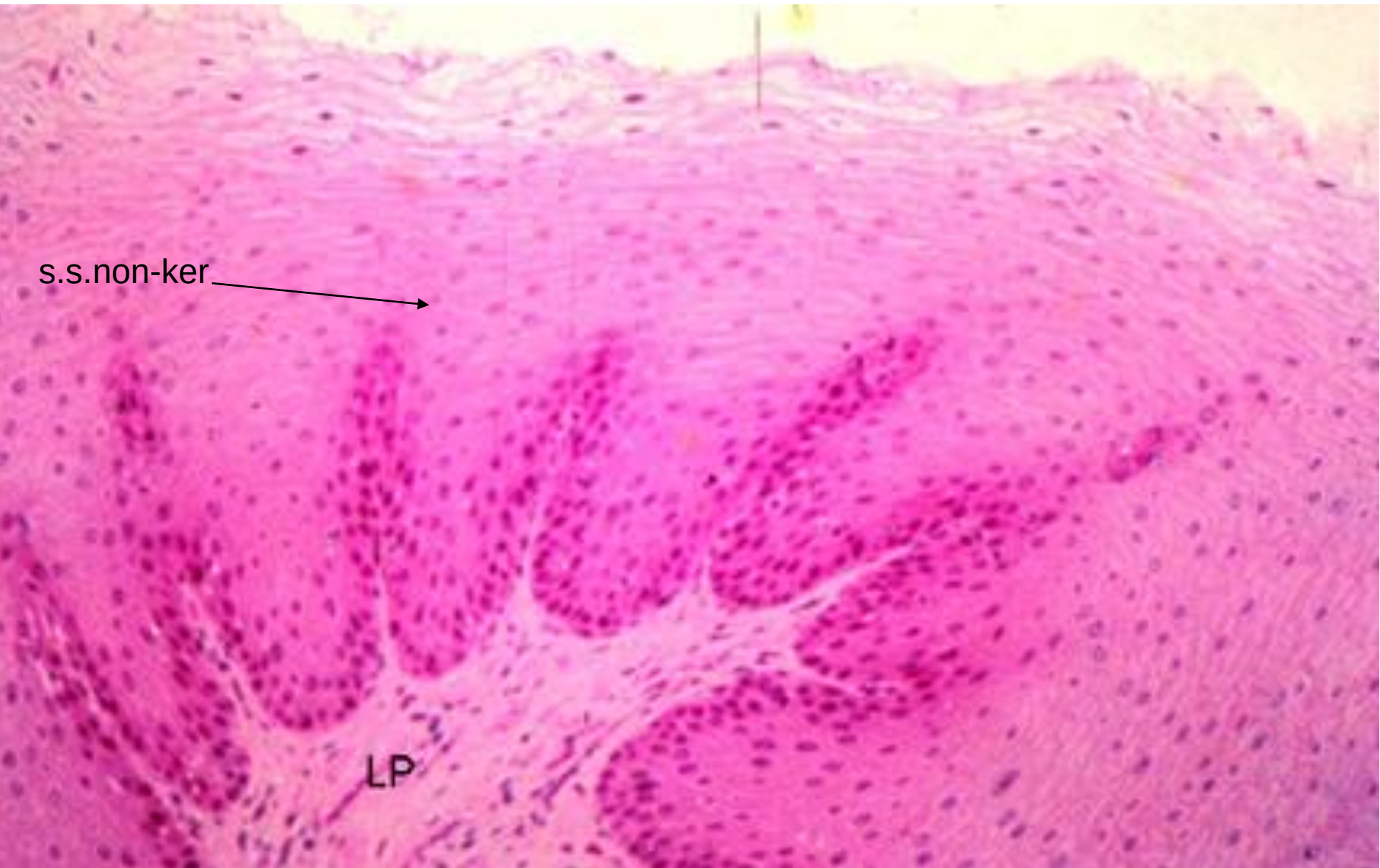
- The esophagus extends from the pharynx to the stomach. Most of the esophagus lies in the thorax and only the last 2-3 cm are within the abdominal cavity . Their wall consist of 4 parts:
 1. **Mucosa**: lined by s.s.ep.non- ker., the lamina propria consist of loose c.t.contain lymphatic nodules , has muscularis mucosa
 2. **Sub-mucosa** : is dense c.t contain mucus g. secrete mucus as a food moves through the esophagus , bundle of elastic & collagen fibers make lumen as stellate shape
 - 3-**Muscularis externa**: is composed of :
 - skeletal muscle** in upper one third ,
 - mixture of skeletal& smooth** in middle one third
 - smooth muscle** in lower one third (**inner circular & outer longitudinal**)
 - 4-**Serosa& adventitia**: only that part of esophagus that is in the peritoneal cavity is covered by **serosa** , the rest is covered by a layer of loose **c.t. the adventitia**



Upper/lower esophageal sphincters



Esophagus



Esophago gastric junction:

is short about (1-1.5 cm.) long of esophagus below the diaphragm , the epithelium change from s. to columnar ep.

in the lamina propria contain group of mucous glands the esophageal cardiac gland that secret mucous

Esophago gastric junction



Thanks